Article

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A Comprehensive Meta-Analysis of Determinants Influencing Military Expenditure: New Methodological Insights and Implications for Defence Budget Allocation

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Abstract: The aim of this article is to conduct a meta-analysis of existing research on the determinants of military expenditure. Using data from an initial screening of 179 studies, 15 studies were selected for meta-analysis, contributing 20,023 observations to analyze key variables influencing defense budget allocations. By employing a random-effects meta-analysis, this study synthesizes findings across diverse geopolitical and economic contexts, addressing inconsistencies in prior research and enhancing the reliability of conclusions. Our findings indicate that war, current military expenditure, and the presence of external threats (enemies) are significant drivers of military spending, while national conditions show a significant negative correlation with military expenditure. Other variables, including GDP, population, democracy, trade, FDI, arms exports, alliances, threats, and political regime type, do not show strong correlations. By combining data from multiple studies, this methodological approach not only generalizes results and improves the precision of effect estimates but also identifies gaps to guide future defense economics research. This research provides policymakers with a broader understanding of the factors shaping defense budgets, offering insights into how traditional budgeting methods might be complemented by a data-driven approach.

Keywords: military expenditure; defence budget allocation; meta-analysis; defence economics; budgetary decision-making

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1 Introduction

The allocation of national defense budgets is a pivotal aspect of government expenditure with profound implications for national security, economic stability, and resource distribution. While previous studies have explored individual factors such as economic, geopolitical, and technological influences, there is a lack of comprehensive meta-analyses that synthesize findings across multiple contexts. This gap limits our ability to draw generalizable conclusions and provide robust guidance for policymakers. This study aims to bridge this gap by using a meta-analysis framework, which integrates findings across diverse contexts to identify consistent trends and reduce variability in conclusions. By applying a random-effects model, this research improves the robustness of estimates and offers a foundation for addressing methodological limitations in previous studies on military expenditure determinants.

In an era characterized by an abundance of research and extensive information, a balanced and critical literature review is crucial for comprehensively assessing the factors impacting defense budget allocation (Stanley 2001).

Despite extensive research identifying the determinants of military expenditure, these studies are frequently limited by their analysis of isolated factors. For instance, studies by Pamp, Dendorfer, and Thurner (2018) have primarily highlighted the impact of GDP and population without incorporating geopolitical factors and national conditions.

Theoretically, defense expenditure is driven by a variety of factors. Economic capacity, typically measured by GDP, is a key determinant, as a stronger economy allows for greater military investment (Goldsmith 2003; Smith 1995). Geopolitical factors, such as alliances and external threats, are frequently cited as central drivers of defense spending, reflecting a nation's security needs (Hartley and Sandler 1995).

This study offers an approach by employing a comprehensive meta-analysis to identify key determinants of military expenditure. Meta-analysis is particularly well-suited for synthesizing diverse research findings, allowing us to generalize results across different contexts and increase the robustness of our conclusions. This method enhances statistical power by combining data from multiple studies, addressing the limitations of individual studies that may focus on isolated factors.

Meta-analysis, as a quantitative literature review method, offers a systematic approach to synthesizing and analyzing the consistency, variation, and effects of previous research findings on the factors influencing military expenditure (Stanley 2001). This method enables the identification of patterns and trends across diverse studies, providing valuable insights into the multifaceted nature of defense budget determinants.

This approach allows for the identification of overarching trends across various studies, increasing the sample size and enhancing the generalizability of findings. Studies were selected based on their methodological rigor and relevance to military expenditure determinants, ensuring that only studies with appropriate statistical data and analysis were included in the meta-analysis.

Meta-analysis is essential for assessing research regarding military expenditure and defense budgets due to its ability to provide a comprehensive and quantitative survey of the existing literature (Alptekin and Levine 2012). While traditional literature reviews may be limited in terms of comprehensiveness and may be dated due to the context within which the data were collected, meta-analysis allows for a more extensive and up-to-date analysis by including a larger sample of estimates from multiple studies (Yesilyurt and Elhorst 2017).

Despite numerous studies on defense spending, many focus on specific variables in isolation. Our meta-analysis addresses this gap by synthesizing multiple variables, including economic, geopolitical, and strategic factors, providing a more comprehensive understanding of the determinants of military expenditure.

This is crucial as it enables a more thorough examination of the effect of military expenditure on economic growth, which is a complex and multifaceted relationship. Additionally, meta-analysis allows for the identification of significant predictors that may not be captured in a single study or systematic literature review (Aveyard and Bradbury-Jones 2019).

2 Materials and Methods

This research, conducted over a three-month period from November 2023 to January 2024, focused on the systematic collection and analysis of studies to identify the determinants influencing military expenditure and the strategies behind defense budget allocations. The first two months (November-December 2023) were dedicated to the literature selection procedure, while January 2024 was used for conducting the meta-analysis and composing the article.

2.1 Literature Selection Procedure

The literature selection procedure began with the identification of studies using relevant keywords. The keywords employed for this study were “Defense Budget,” “Military Expenditure,” “Defense budgeting,” and “Determinants of Military Expenditure.” To conduct our research, we exclusively utilized the Scopus Elsevier database. We selected Scopus due to its extensive collection of abstracts, citations, and scholarly articles, which enhances the reliability and accessibility of our findings.

We acknowledge that limiting the study to Scopus-indexed publications may introduce publication bias, as studies with insignificant or non-confirmatory results may be less likely to be published. To address this limitation, future research could include grey literature, such as unpublished working papers, to provide a more comprehensive view and further mitigate potential bias.

The decision to use Scopus was based on its focused coverage, distinguishing it from broader tools like Google Scholar. Notably, Scopus is particularly beneficial for humanities and social sciences, offering advantages over databases such as ScienceDirect or Web of Science. Additionally, Scopus is recognized for its repository of high-quality peer-reviewed articles and advanced filtering options (Martín-Martín et al. 2021).

2.2 Visualization of Keywords

In the field of defence budget research, as depicted in Figure 1, "defence spending," "military expenditure," "military spending," and "defence economics" are the most frequently used keywords. Military expenditure is a topic that necessitates in-depth study, particularly in relation to its impact and association with economic factors and defence policies. This analysis illustrates the interactions between defence budget allocations and various economic and industrial management aspects, highlighting the importance of comprehensively understanding how military spending is structured and its influence on and by global socio-political conditions.

Based on the analysis using VOSviewer, military expenditure and military spending were widely used keywords in articles concerning defence budgeting over the last two decades (Figure 2). This indicates the opportunity for further studies related to these keywords.

2.3 Period of Literature

The period of the literature was 2010–2023. This timeframe captures a significant era of global geopolitical shifts, including rising tensions in various regions, the emergence of new military technologies, and changes in international alliances. The period post-2010 is marked by the recovery phase of the global economy following the 2008 financial crisis. This context is crucial for evaluating how economic recoveries and downturns impact defence budget allocations.

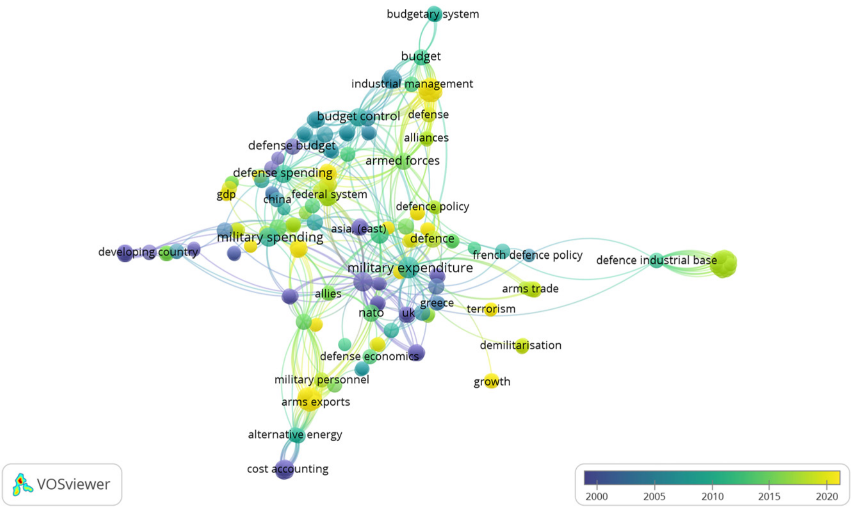

Figure 1: Overlay visualization of articles dealing with defence budget research (developed by the authors).

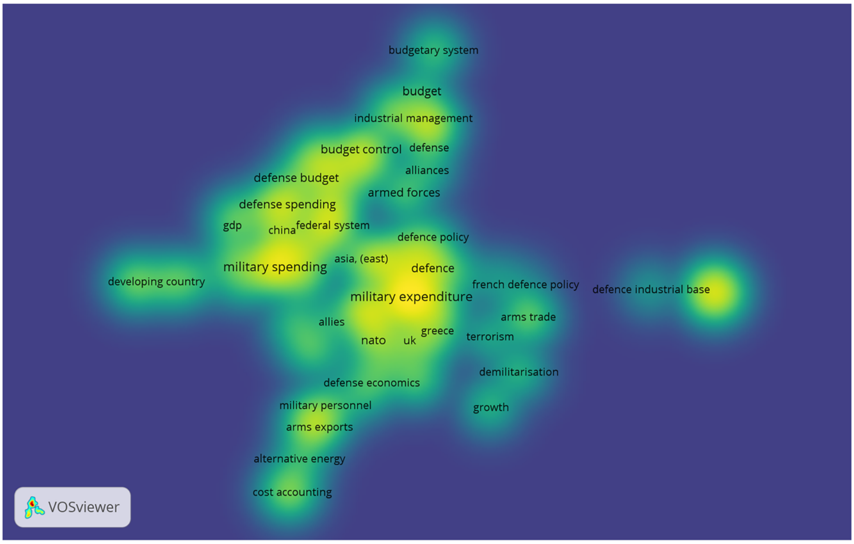

Figure 2: Visualization of keywords density (developed by the authors).

2.4 Literature Selection Process

Figure 3 showed the literature selection process comprised four distinct procedures: Identification, Screening, Eligibility, and Inclusion. In the Identification phase, we conducted a comprehensive search using the Scopus database with specific keywords related to military expenditure, such as “military expenditure,” “defense budget,” and “determinants of defense spending.” This resulted in an initial pool of 428 studies. During the Screening phase, we reviewed the titles and abstracts of these studies to assess their relevance to the research objectives. Only studies that explicitly examined the determinants of military expenditure were selected, narrowing the pool to 179 relevant studies. In the Eligibility phase, we thoroughly reviewed the full text of the remaining 76 studies. We applied strict criteria to determine whether each study was suitable for inclusion in the meta-analysis. First, we selected studies that provided empirical data with detailed statistical results, such as effect sizes, regression coefficients, or correlations, ensuring comparability across multiple studies. Second, we included studies that analyzed key variables relevant to military expenditure (e.g. GDP, population, war, democracy, military alliances). These variables had to be presented in a standardized format, allowing for synthesis and comparison with other studies. We only included studies that employed robust

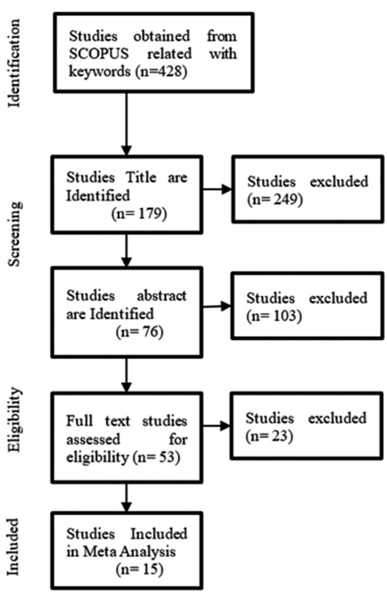

Figure 3: The process of literature selection (developed by the authors).

quantitative methods, such as regression analysis, panel data models, or similar statistical approaches, to maintain methodological consistency. Additionally, we required studies to report sufficient sample sizes (N) for accurate effect size calculations. Finally, we included only peer-reviewed journal articles to ensure the reliability and quality of the data. We excluded studies that were purely theoretical, lacked the necessary statistical information, or did not meet these inclusion criteria.

In the Inclusion phase, we selected 15 studies that met all the eligibility criteria for the meta-analysis. These studies provided the necessary data for a comprehensive synthesis of the determinants influencing military expenditure (Figure 3).

2.5 Meta-Analysis Methodology

Table 1 shows the 15 selected studies, outlining the authors, sample sizes (N), dependent and independent variables, and methodologies. We analyzed variables that appeared in at least two different studies. This meta-analysis includes a total of 20,023 samples, with each study contributing a different number of data points. In this analysis, 'samples' refer to the independent data entries we collected from each study, representing the individual observations or datasets used. The definitions of these chosen variables are provided in Table 2.

The meta-analysis included 15 studies, covering a total of 20,023 observations. Independent variables utilized in this study included population, war/threats, enemies, alliances, democracy, GDP, political regime, previous year military expenditure, arms export, and trade. Variables were selected from studies that discussed them in at least two different sources, as a minimum of two studies is required to perform a meta-analysis (Valentine, Pigott, and Rothstein 2010). The assumption that the studies are conducted on varying populations is largely true, as the studies in our analysis span different countries, time periods, and contexts. This variability is accounted for in the random-effects model, which allows for heterogeneity across studies. The definitions, measurements, hypotheses, and sources of these variables are detailed in Table 2.

To convert to Pearson's correlation coefficient (Pearson-r), the Z score method was employed, involving the calculation of the Z score using the probability levels (p-values) from the 15 earlier studies. First Z-scores were calculated from the reported p-values in each study. The Z-scores were then converted to Pearson's r using the following formula:

$$\text{Pearson}\, r = \frac{Z}{\sqrt{N}} \tag{1}$$

where Z is the Z score and N is the sample size of the study. Pearson-r serves as the

Table 1: Studies are used in this research.

No	Authors	Type	Scope	Observation	Y	X	Method
1	Bove and Nisticò (2014)	Article	World wide	135 countries	Military burden (as a share of GDP)	Population, alliances, military regime, military participation in politics, regional military spending	Pooled OLS, panel data with fixed effects and IV estimates.
2	Hou, D. (2018)	Article	World wide	29 countries Asia and oceania	Annual military expenditure (ME)	GDP, population, military expenditure, trade openness	Dynamic panel approach
3	Pamp and Thurner (2017)	Article	World wide	159 country	Logmilexp (e logarithm of military expenditures measured in million)	Military expenditure, arms export, GDP, interstate condition, foes, friends, population	Dynamic panel data models
4	Do (2021)	Article	World wide	163 countries	Military expenditure	Natural resources rent, income, GDP growth, trade, political stability, corruption control, terror attacks, previous year military expenditure	Dynamic panel approach
5	Deng and Sun (2017)	Article	World wide	95 countries	Military expenditure	State executive, state legislative, GDP, war, alliance, polity, year	Dynamic panel approach
6	Kollias et al. (2018)	Article	Russia	100 data set	Russian military spending (RUSM)	GDP, population, oil prices, liberal democratic party	Ordinary least squares (OLS) and generalized additive mixed models (GAMMS)
7	Töngür and Elveren (2015)	Article	World wide	134 countries	Military expenditures as a percentage of GDP	Previous year military expenditure, foes, friends, war, economic growth	Dynamic panel approach
8	Bove and Brauner (2016)	Article	World wide	112 countries	Log military burden	Democracy, interstate war, population, trade openness	Dynamic panel data analysis.
9	Graham and Mueller (2019)	Article	OECD countries	33 countries	Household income inequality (EHII)	Military expenditure, FDI, social welfare, GDP growth, labor force, population, government consumption	Dynamic panel data analysis
10	Gibson (2020)	Article		563 observation			

Table 1: (continued)

No	Authors	Type	Scope	Observation	Y	X	Method
			Developing countries		Military spending as a share of GDP	FDI, GDP, state capacity, democracy, population, logged GCC rentier states	Pooled cross-sectional approach
11	Pamp, Dendofler, F. and Thurner (2018).	Article	World wide	2,227 observation	Log of military expenditure	Military expenditure, population, export, GDP, interstate conflict	Dynamic panel data analysis
12	Saba and Ngepah (2019)	Article	African countries	35 countries in africa	Manufacturing value-added as a percentage share of the GDP	GDP, military expenditure, human capital, population, import of goods and export of goods	Recently developed panel causality and generalized method of moments (GMM)
13	Armey and Mcnab (2017)	Article	World wide	3,598 observation	Military share	Civil war, war, previous year military share	Ordinary least squares (OLS) and generalized method of moments (GMM)
14	Markowski, Chand and Wylie (2017)	Article	Indo pacific	16 countries	Military expenditure and GDP by country ‘j’ at time ‘t’	Natural logarithms of military expenditure (i.e. Lnmilex), GDP, and population	Regression & ARDL
15	Vallejo-Rosero et al. (2021)	Article	Nato countries	27 countries	Military expenditure, human development index (HDI), global peace index (GPI)	Health expenditure, education expenditure, GDP, R&D cost,	Dynamic panel data models

Table 2: Definition of variables.

Variables	Definition	Author
Population	Countries population size	(Bove and Brauner 2016; Bove and Nisticò 2014; Deng and Sun 2017; Gibson 2020; Graham and Mueller 2019; Hou 2018; Pamp and Thurner 2017; Saba and Ngepah 2019)
War	1 if there is a war of any kind; =0 otherwise	(Armey and McNab 2017; Bove and Brauner 2016; Deng and Sun 2017; Töngür and Elveren 2015)
Allies/friends	ln weighted spending of friends/allies	(Bove and Nisticò 2014; Deng and Sun 2017; Pamp and Thurner 2017; Töngür and Elveren 2015)
Arms export	Arms exports in mill.TIV	(Pamp and Thurner 2017; Pamp, Dendorfer, and Thurner 2018; Töngür and Elveren 2015)
Democracy	Polity IV index score ranging from −10 (autocratic) to 10 (democratic)	(Bove and Brauner 2016; Gibson 2020; Kollias et al. 2018)
Foes	ln weighted spending of foes	(Pamp and Thurner 2017; Töngür and Elveren 2015)
GDP	Countries GDP percapita and growth	(Deng and Sun 2017; Do 2021; Graham and Mueller 2019; Gibson 2020; Hou 2018; Kollias et al. 2018; Markowski, Chand, and Wylie 2017; Pamp, Dendorfer, and Thurner 2018; Pamp and Thurner 2017; Saba and Ngepah 2019; Töngür and Elveren 2015; Vallejo-Rosero et al. 2021)
National condition	Constructed from a collection of data such as state executive, state legislative, state capacity, government consumption, and militarized interstate disputes onset. Refers to the internal conditions and capabilities of a country that influence policy and military expenditure.	(Deng and Sun 2017; Graham and Mueller 2019)
Political regime	Degree of military participation in politics.	(Bove and Nisticò 2014; Deng and Sun 2017; Do 2021)
Threats	Threats variable, constructed from a collection of data such as terror attacks, interstate conflict, and regional effects of unrest, refers to various forms of security threats faced by a country.	(Do 2021; Gibson 2020; Pamp and Thurner 2017; Pamp, Dendorfer, and Thurner 2018)
Trade	log trade (% of GDP)	(Do 2021; Hou 2018)
FDI	FDI out. % Of GDP	(Gibson 2020; Graham and Mueller 2019)
Military expenditure	Previous year military expenditure as % of GDP	(Bove and Nisticò 2014; Do 2021; Pamp and Thurner 2017; Pamp, Dendorfer, and Thurner 2018)

measure of effect size, calculated using N and Pearson-r values, followed by a standard meta-analysis employing the random-effects model. The random-effects model because it accounts for variability across different studies, which is particularly useful when studies include diverse populations or contexts. The use of the random-effects meta-analysis in this study addresses key limitations in previous single-study analyses by accounting for heterogeneity across studies. This approach ensures that findings are not overly influenced by context-specific conditions, allowing for more generalizable conclusions. Additionally, the weighting mechanism within the model enhances the reliability of effect size estimates by balancing contributions from studies of varying sample sizes and methodological rigor. By synthesizing diverse research contexts, this method provides a robust platform for identifying overarching trends and informing more nuanced future analyses. Weights (W_i) were calculated for each study based on its sample size and variance, and the combined effect size (Y_i) was determined by averaging the weighted Pearson r values across all studies. This method allows us to generalize findings while accounting for differences in study designs and populations. (Borenstein, Hedges, and Rothstein 2007)

$$M* = \frac{\sum_{i=1}^{k} W_i^* Y_i}{\sum_{i=1}^{k} W_i^*} \tag{2}$$

The random-effects model also accommodates for study-level variability, ensuring that the average effect size reflects the general trend while adjusting for heterogeneity. This variability, or random error, could result from differences in study methodologies, sample characteristics, or geopolitical contexts. By applying this model, we provide a robust estimate of the overall effect size across different studies. All these results are then analysed using standard meta-analysis techniques in OpenMEE, which is a software specifically designed for conducting meta-analyses.

3 Results

The meta-analysis of the 15 selected studies reveals several factors that determine military expenditure. This research highlights military expenditure as the dependent variable, while population, war, allies, arms export, democracy, enemies, GDP, national condition, political regime, threats, trade, FDI, and previous year's military expenditure are considered independent variables.

The analysis shows a statistically significant positive relationship between military expenditure and variables such as war, foes, and previous year's military expenditure. This result align with prior meta-analyses on military expenditure, such as those conducted by Alptekin and Levine (2012), which also identified war and external threats as significant determinants.

This indicates that countries facing conflict or having a high level of military spending in the past tend to allocate more resources to their defense budgets. Conversely, there is a significant negative relationship between military expenditure and national conditions, suggesting that better internal conditions might reduce the need for higher military spending.

The results provide a broader scope by including variables such as national conditions and political regime types, which were less emphasized in previous studies. Moreover, while prior meta-analyses have found a strong correlation between GDP and military spending (Pamp and Thurner 2017), our study finds a weaker relationship. This divergence may be due to the inclusion of more recent studies and data or differences in the geopolitical contexts examined (Table 3).

Variables such as population, democracy, allies, GDP, political regime, threats, trade, FDI, and arms export did not show a statistically significant impact on military expenditure in our analysis.

The significant positive relationship between military expenditure and war supports findings by Christie (2017) and Skogstad (2015), highlighting that countries increase their defence budgets in response to regional instability. Similarly, the influence of previous year's military spending on current expenditure underscores the importance of historical spending patterns, as noted by Do (2021), Hou and Chi (2021), and Pamp and Thurner (2017).

Table 3: The estimation of meta-analysis: determinant of military expenditure.

Variables	Studies	Estimate	Standard errors	P-Value
Population	9	0.009	0.026	0.739
War	**4**	**0.048**	**0.011**	**<0.001[c]**
Allies	4	−0.003	0.036	0.942
Arms export	4	−0.019	0.044	0.665
Democracy	3	−0.001	0.075	0.987
Foes	**2**	**0.052**	**0.015**	**<0.001[c]**
GDP	14	0.008	0.018	0.634
National condition	**5**	**−0.049**	**0.02**	**0.015[b]**
Political regime	4	0.018	0.038	0.63
Threats	4	0.028	0.035	0.424
Trade	2	0.018	0.06	0.764
FDI	2	−0.02	0.153	0.895
Military expenditure	**4**	**0.05**	**0.012**	**<0.001[c]**

[a], [b]and [c] = significant at 10 %, 5 % and 1%.

The insignificant influence of international relations dynamics, such as threats and regional interactions with allies, diverges from studies by Christie (2017) and Yesilyurt and Elhorst (2017), which suggest these dynamics play a significant role in shaping defence budgets. This discrepancy highlights the complex and multifaceted nature of defence spending determinants.

Additionally, the lack of a strong relationship between GDP and military expenditure challenges traditional economic models, as suggested by Pamp and Thurner (2017), Solarin (2017), and Vallejo-Rosero et al. (2021). This indicates that economic prosperity alone does not dictate military spending levels.

The negative association between national conditions and military expenditure aligns with Do (2021) and Bakirtas and Akpolat (2020), indicating that countries with substantial internal stability or resource wealth might prioritize domestic development over military expansion.

The analysis also challenges the idea that political regime types directly influence military expenditure, contrary to findings by Bove and Brauner (2016) and Yesilyurt and Elhorst (2017). This suggests that the impact of regime type on military spending is influenced by a range of factors beyond a simple democratic versus non-democratic dichotomy.

Lastly, the insignificant relationship between population size and military expenditure contradicts the findings of Pamp and Thurner (2017) and Skogstad (2015), implying that demographic factors may not directly affect defense budgets. Instead, other elements such as strategic priorities and external threats play a more critical role.

4 Discussion

The findings from this meta-analysis provide valuable insights into the determinants of military expenditure, challenging some traditional views and supporting others.

4.1 Impact of War and Enemies on Military Spending

The significant positive relationship between military expenditure and war highlights the direct impact of conflict on defense budgets. Countries facing regional instability or direct threats from enemies tend to allocate more resources to their military, reflecting a security dilemma where states increase their military capabilities in response to perceived threats. This aligns with the findings of Christie (2017) and Skogstad (2015), reinforcing the centrality of security concerns in defense budget decisions.

4.2 Historical Spending Patterns

The influence of previous year's military expenditure on current spending underscores the importance of historical spending patterns. Defense budgets are not solely reactive to present conditions but are also shaped by past financial commitments and strategic choices. This is consistent with the observations of Do (2021), Hou and Chi (2021), and Pamp and Thurner (2017), indicating that defense budgeting practices often rely on continuity and past trends.

4.3 Internal Conditions and Defense Budgets

The negative correlation between national conditions and military expenditure suggests that internal stability and favorable conditions may reduce the need for higher defense spending. This finding supports the notion that countries with better internal governance and stability might prioritize domestic development over military expansion. Similar conclusions were drawn by Do (2021) and Bakirtas and Akpolat (2020), highlighting the complex interplay between domestic conditions and defense spending priorities.

4.4 Economic Factors and Defense Spending

Contrary to some traditional economic models, this study finds that GDP does not have a strong correlation with military expenditure. This challenges the assumption that economic prosperity directly influences defense budgets. Instead, the relationship between economic growth and military spending appears to be more nuanced, suggesting that other factors, such as strategic priorities and geopolitical considerations, play a more significant role. This finding diverges from the research by Pamp and Thurner (2017) and Vallejo-Rosero et al. (2021), necessitating a reevaluation of economic determinants in defense budgeting.

4.5 Geopolitical and International Relations Factors

The insignificant influence of variables like trade, FDI, and alliances on military expenditure highlights the complex nature of defense budget determinants. While previous studies, such as those by Christie (2017) and Yesilyurt and Elhorst (2017), suggested that geopolitical dynamics significantly impact defense budgets, this analysis indicates that such influences may not be as straightforward. The role of international relations in shaping military spending appears to be contingent on various contextual factors, necessitating further investigation.

4.6 Political Regime and Military Spending

The analysis indicates that the relationship between political regimes and military expenditure is not straightforward. While some studies, like those by Bove and Brauner (2016) and Yesilyurt and Elhorst (2017), linked higher military spending with autocratic regimes, our findings suggest that this relationship is influenced by multiple factors beyond the simple democratic versus non-democratic dichotomy. This complexity underscores the need for a more nuanced understanding of how political contexts shape defense budgets.

4.7 Population Size

The finding that population size does not significantly impact military expenditure challenges the notion that larger populations directly correlate with higher defense spending. This contradicts previous research by Pamp and Thurner (2017) and Skogstad (2015), suggesting that demographic factors might be less critical than previously thought. Instead, strategic priorities and external threats may have a more substantial influence on military budgets.

5 Conclusions

This meta-analysis offers a comprehensive understanding of the determinants of military expenditure, emphasizing the importance of security concerns and historical spending patterns. The findings emphasize the need for policymakers to adopt a balanced, data-driven approach to defense budgeting by leveraging historical spending trends as a baseline while integrating real-time geopolitical and security indicators. The strong influence of external threats, such as war, highlights the necessity of dynamic budget allocation systems that respond swiftly to regional instability. Moreover, the negative correlation between domestic stability and military expenditure underlines the importance of strengthening governance and economic resilience to reduce excessive defense spending and reallocate resources toward developmental priorities. These insights offer a strategic framework for optimizing resource allocation in evolving security landscapes.

6 Future Research Direction

While this study provides significant insights, it also highlights gaps that warrant further investigation. Future research should explore the evolving impact of geopolitical tensions, including cyber threats and international alliances, on military spending. Additionally, examining the influence of internal factors such as political stability and institutional quality on defence budgets can enhance our understanding of defence economics. While our primary focus was on economic and geopolitical factors, we acknowledge the importance of government ideology, cyber threats and international alliances, in shaping military expenditure.

In conclusion, this research serves as a practical tool for policymakers, offering an alternative to relying solely on previous years' defence budget data. By incorporating a broader set of determinants, defence budgeting can better address contemporary security challenges and strategic priorities.

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